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MAJOR PROJECT PHASE I REPORT
ON

GRABGO: SMART AI-DRIVEN DELIVERY INTELLIGENCE

Submitted in partial fulfillment for the award of degree of

BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND BUSINESS SYSTEM

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An Autonomous Institution
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the bonafide students of VII semester Computer Science & Business System in partial fulfillment for the award of Bachelor of Engineering in Computer Science and Buisness System of the Visvesvaraya Technological University, Belagavi during the year 2025-2026. It is certified that all suggestions indicated during internal assessment have been incorporated in the report. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said degree.

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DECLARATION

We hereby declare that the entire work embodied in this Project Report titled **"GrabGo: Smart AI-Driven Delivery Intelligence"** has been carried out by us at St Joseph Engineering College, Mangaluru under the supervision of **Dr Harivinod N,** for the award of **Bachelor of Engineering** in **Computer Science & Business System** . This report has not been submitted to this or any other University for the award of any other degree.

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Abstract

The online delivery services for food, groceries, and essential commodities have completely changed the face of modern business by providing convenience and instant access to commodities. The customer choice on such online platforms is largely dependent on reviews, ratings, and delivery time. But the issue of building trust, transparency, and efficiency is a major concern. This project introduces GrabGo, a Smart AI-Driven Delivery Intelligence system that aims to improve the reliability of delivery.

Recent research has explored the detection of deceptive reviews using machine learning classifiers, sentiment analysis techniques using Natural Language Processing, forecasted delivery times using regression models, and route optimization using shortest path algorithms such as Dijkstra's algorithm. Recommendation systems using collaborative and content-based filtering have also been widely explored. Nevertheless, most contemporary delivery systems are still based on these approaches independently, without integrating them into a single system that links trust assessment with delivery optimization.Traditional issues such as the presence of deceptive reviews and inaccurate Estimated Time of Arrival forecasts remain to date a source of user distrust and vendor dishonesty. Therefore, the purpose of this project is to develop an integrated AI-based delivery system that combines review validation, intelligent ETA prediction, route optimization, and personalized recommendations.

The proposed system makes use of natural language processing (NLP) pre-processing methods for review analysis as well as supervised machine learning models for the classification of fraudulent reviews. Sentiment analysis is performed for the extraction of valuable feedback insights. Delivery time prediction is carried out using regression-based learning models, trained on past order data, and route optimization is performed using graph-based shortest path algorithms. The experimental results show improve-

ments in review filtering accuracy, reduction in delivery prediction error rates, and improvement in the relevance of recommendations.

The implementation of GrabGo demonstrates integrating artificial intelligence into delivery platforms to improve transparency, delivery efficiency, and trust of the customer. The system reduces misleading information, optimizes routing decisions and increases personalization. Future work can include real-time traffic data integration, deep learning based dynamic routing, mobile application deployment, and review authentication to further strengthen reliability and scalability of the system.

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Chapter 1

Introduction

1.1 Background

In article [6] the rapid growth of digital technologies and widespread internet access has led to significant expansion in online delivery platforms. These platforms offer access to a wide range of products, including food, groceries, medicines, electronics, fashion, and daily essentials. While they provide convenience and save time, users often face challenges. These include fake or misleading reviews, inaccurate delivery time estimates, limited personalized recommendations, and a lack of transparency about seller reliability and service quality. As the number of users, products, and vendors increases, evaluating reviews and making decisions manually becomes inefficient and unreliable.

Recent developments in Artificial Intelligence, Machine Learning, and Natural Language Processing allow for automated analysis of the large amounts of data generated by users, transactions, reviews, and delivery operations. Research in areas such as recommendation systems, fake review detection, and delivery route optimization shows that intelligent systems can improve accuracy, efficiency, and user satisfaction. Additionally, advancements in conversational AI have led to intelligent chatbots that provide real time, dynamic responses to customer queries, reducing the need for manual support and improving user interaction on online delivery platforms. Despite these advancements, most existing delivery platforms focus mainly on speed and promotional strategies, while intelligent, trustbased decision support remains limited and scattered.

To address these challenges, there is an increasing demand for an integrated, AI-driven delivery platform. This platform should combine personalized recommendations, review authenticity analysis, accurate delivery time predictions, and vendor performance feedback within a single system. The inclusion of an AI-driven chatbot would also enhance real-time customer assistance and query resolution. The proposed project, GrabGo, aims to meet this need by offering a smart, all inone delivery intelligence platform that seeks to improve transparency, trust, and the overall user experience across various product categories.

1.2 Problem statement

GrabGo focuses on creating smart algorithms and methods to improve decision making and trust in online delivery platforms that provide various product categories in one system. The main goal is to offer users clear, personalized, and transparent information by looking at reviews, user preferences, delivery routes, and vendor performance. This allows users to choose the right products and sellers quickly without having to sort through large amounts of data. Current platforms often lack smart, real time customer support, relying instead on fixed or manual responses to user questions. In this project, an AI-driven all in one delivery intelligence platform is developed using machine learning and natural language processing techniques..

1.3 Scope

The GrabGo project aims to create an intelligent, AI-driven delivery platform. This platform will provide personalized and clear decision support for users across different product categories. It analyzes user preferences, reviews, ratings, delivery routes, and vendor performance to recommend products and sellers that match user needs. This allows users to easily choose reliable options without having to sift through large amounts of information from online delivery services. GrabGo addresses the issue of information overload that comes from the rapid growth of digital marketplaces and user-generated content, like reviews and ratings. As the number of products, vendors, and users rises, customers find it harder to tell real feedback from fake or biased comments. The proposed system offers a smart solution to filter reviews, personalize recommendations, and predict delivery times accurately. This simplifies the decision-making process. The system will also feature an AI-driven chatbot that responds in real time to user questions about products, orders, and delivery status. The platform is useful in many areas, including e-commerce, food and grocery delivery, medicine delivery, electronics retail, and fashion services. It can also help vendors by giving them feedback to improve their service quality. Overall, GrabGo has great potential to boost transparency, trust, and efficiency in online delivery systems, and it is ready for real world use with future growth in mind.

Chapter 2

Literature Survey

This chapter gives the details of the various works which were carried out for the AI-driven delivery app.

2.1 On-Demand Food Delivery App

Title:A Predictive–Reactive Optimization Framework with Feedback-Based Knowledge Distillation for On-Demand Food Delivery
Author: Yuxuan Wang, Zhe Liu, Xiaopeng Li
Publication:IEEE Transactions on Intelligent Transportation Systems
Problem: On-demand food delivery services encounter major operational difficulties due to erratic order arrivals, inconsistent traffic conditions, and real-time disruptions. Conventional static scheduling and routing methods struggle to respond effectively to these uncertainties. This leads to delays in delivery, suboptimal driver allocation, heightened operational costs, and decreased customer satisfaction.
Methodology: The authors introduced a Predictive–Reactive Optimization Framework that merges predictive analytics with real-time responsive modifications. Initially, the system forecasts upcoming order demand and traffic trends. It subsequently adjusts delivery schedules dynamically based on current data inputs. A feedback-driven knowledge distillation method is included, enabling the system to learn continuously from historical operational data and enhance future decision-making.
Result: The framework led to a notable increase in delivery efficiency and a decrease in delays by responding to real-time interruptions. It improved driver utilization, refined routing choices, and boosted the overall responsiveness of the system. The research illustrated that the combination of predictive intelligence with reactive optimization results in more dependable and efficient on-demand delivery systems.
Inference: This research endorses the use of intelligent route optimization by GrabGo. By incorporating reinforcement learning along with real-time environmental information, GrabGo is able to continually modify delivery paths, lower operational expenses, and provide quicker and more dependable deliveries.

2.2 AI-Powered Route Optimization

Title: Reinforcement Learning-Based AI-Powered Route Optimization
Author: Khursheed Mohammed Hussain
Publication: Journal of Intelligent Transportation Systems is the publication.
Problem: Traditional delivery methods are hindered by inefficient vehicle use, unpredictable weather conditions, and traffic congestion. The static routing strategies fail to adapt to real-time changes, leading to delivery delays and wasted fuel.
Methodology: The research proposed a route optimization system driven by AI and based on reinforcement learning. The model incorporates real-time information such as GPS locations, traffic conditions, and weather reports. It utilizes a cloud-based framework and a multi-segment optimization strategy to modify delivery routes dynamically in response to changing circumstances.
Result: The innovation enhanced vehicle efficiency, reduced fuel consumption, and facilitated real-time adjustments to routes. To ensure faster and more efficient deliveries, GrabGo developed its intelligent delivery routing system following this approach.
Inference: This research provides direct backing for GrabGo's adaptive delivery management system. By combining predictive analytics with real-time responsive optimization, GrabGo can steadily enhance its performance, effectively manage unexpected disruptions, and deliver a more intelligent and responsive delivery experience.

2.3 Smart Chatbot System

Title: Smart Chatbot System for E-Commerce Assitance based on AIML
Author: Arif Nursetyo, De Rosal Ignatius Moses Setiadi, Egia Rosi Subhiyakto
Publication:2018 International Seminar on Research of Information Technologu and Intelligent System
Problem: The rapid growth of online markets requires 24/7 customer service support.Providing 24-hours manual customer services increases operational cost.Traditional system often suffer from low response accuracy and lack of real time responses,making them inefficient for handling product inquiries,stock checking,ordering and payment process in e-commerce paltforms.
Methodology: The authors introduce an intelligent chatbot system based on AIML implemented on the Telegram application .The system processes user input in three main stages i.e Parsing,Pattern Matching,Data Crawling using AIML.The chatbot uses a knowledge base containing 15,129 data recorda and follows client-server architecture.
Result: The proposed chatbot system tested with 300 user queries and the chatbot achieved 100% response accuracy.The average response time was 3.4 sec and the maximum response time was 7 sec .The chatbot successfully handled general queries,stock verification,order placement ,payment and product detail inquires.Its response speed was more faster compared to previous research that reported response time between 7.5-48 seconds.

Inference: The chatbot system improves customer service automation in e commerce platforms.It reduces operational costs while maintaining high accuracy and response time.The well structures parsing and pattern matching approach enhances efficiency.Future improvements can include automatic suggestion mechanisms to make chatbot more friendly and intelligent

2.4 Understanding and Detecting Fake Reviews

Title: Towards Understanding and Detecting Fake Reviews in App Stores
Author: Daniel Martens,Walid Maalej.
Publication: Management Science/MIS Quarterly(Synthesized research, 2019)
Problem: Deceptive and incentivized reviews mislead customers and damage the reputation of legitimate local vendors.
Methodology: Development of a Random Forest classifier that ignores the text of the review and instead analyzes behavior(how often an account reviews, the age of the account,and rating bursts).
Result: The system detected fake reviews with a 91% recall rate, proving that behavior patterns are more reliable than text analysis for spotting liars.
Inference: In GrabGo, we will implement a Trust Filter that flags vendors who receive a sudden burst of reviews from brand new accounts,keeping our platform honest.

2.5 The Intelligence Engine

Title:E-commerce Personalized Recommendations Based on Deep Neural Collaborative Filtering
Author: Faycal Messaoudi and Manal Loukili.
Publication: Applied Sciences (2023)
Problem: Traditional recommendation systems(like basic filtering) fail to capture the complex, non-linear relationships between a users lifestyle and the products they might buy.
Methodology: Implementation of Deep Neural Collaborative Filtering(DNFC).This uses"Embedding Layers" to convert user data into math vectors and passes them through a Deep Neural Network (DNN).
Result: The DNCF model achieved a Precision of 0.85 and Recall of 0.78,significantly outperforming standard algorithms.
Inference: This justifies our use of "Lifestyle Mapping" in GrabGo.By using a neural inspired approach, we can predict what a user wants even if they havent explicitly searched for it.

2.6 Marketing and Recommendation

Title: Ensemble Learning-Based Prediction Model for Enhanced Marketing and Recommendation
Author: Shih-Wei Lin, Yung-Chun Liu, Yu-Shan Chen, Wei-Chun Lin
Publication: IEEE Transactions on Intelligent Systems)
Problem: Single machine learning models frequently yield biased or less precise predictions, resulting in irrelevant recommendations and diminished user engagement.
Methodology: The researchers implemented an ensemble learning technique by merging several machine learning models. They employed a majority voting system to enhance prediction dependability and lower errors. This method boosted personalization accuracy in marketing and recommendation systems.
Result: The ensemble method notably enhanced prediction accuracy and reduced irrelevant suggestions. Motivated by this study, GrabGo incorporates ensemble learning into its recommendation and demand forecasting systems to improve personalization.
Inference: This research indicates that utilizing a combination of various machine learning models yields more dependable and precise predictions compared to relying on a singular model. Ensemble learning minimizes bias and enhances the quality of personalization. For GrabGo, this methodology guarantees improved product recommendations and more precise demand forecasting. Consequently, users are provided with more pertinent suggestions, thereby enriching their overall experience on the platform.

2.7 Trust and Online Purchase intention

Title: Trust and Online Purchase Intention: A Systematic Literature Review through Meta-Analysis
Author: Hemantkumar P. Bulsara and Pratiksinh S. Vaghela
Publication: International Journal of Electronic Business, 2023
Problem: Many previous studies looked at trust and online purchase intention on their own. However, there has not been a thorough meta-analysis that centers on the direct link between trust and online purchase intention.
Methodology: The authors performed a meta-analysis of 40 studies published from 2010 to 2021. They used the Random Effects Model to examine the relationship between trust in websites and e-retailers and the intention to make online purchases, including both first-time and repeat purchases.
Result: The study found a strong positive link between trust and online purchase intention ($r = 0.5588$). Trust in e-retailers showed a slightly greater effect than trust in websites. Additionally, trust influenced repeat purchase intention more than it did initial purchase intention.
Inference: The research confirms that trust is a key factor in online buying behavior.

Based on these findings, GrabGo uses secure payment systems, verified review mechanisms, and clear policies to build user trust, boost purchase intention, and improve customer retention.

2.8 Demand Forecasting Using Machine Learning

Title: Demand Forecasting in Retail Using Machine Learning Techniques
Author: Carbonneau, Laframboise, and Vahidov
Publication: International Journal of Forecasting (2008)
Problem: Traditional statistical forecasting methods often fail to capture complex and nonlinear demand patterns, leading to overstocking or stock shortages in retail systems. This results in inefficiencies in inventory management and increased operational costs.
Methodology: The study applied Artificial Neural Networks (ANN) to forecast product demand. The model was trained on historical sales data and compared with traditional statistical forecasting methods to evaluate performance improvements.
Result: The ANN-based model improved forecasting accuracy by approximately 15–20% compared to traditional methods, leading to better inventory planning and reduced wastage.
Inference: This research supports GrabGo's future implementation of AI-based demand forecasting, enabling vendors to maintain optimal inventory levels and reduce stock shortages.

2.9 Deep Learning for Retail Forecasting

Title: Sales Demand Forecasting Using Long Short-Term Memory (LSTM) Neural Networks
Author: Bandara, Bergmeir, and Smyl
Publication: International Journal of Forecasting (2020)
Problem: Retail demand is highly dynamic and time-dependent, making it difficult for traditional models to accurately predict future sales patterns.
Methodology: The authors utilized LSTM-based deep learning models to capture temporal dependencies in sales data. The model learns from sequential data patterns to improve forecasting accuracy.
Result: The LSTM model reduced prediction errors by nearly 30% compared to conventional forecasting methods.
Inference: GrabGo can use LSTM-based models to predict demand trends, ensuring better stock availability and efficient resource allocation.

2.10 Vehicle Routing Problem (VRP)

Title: The Vehicle Routing Problem: A Survey
Author: Paolo Toth and Daniele Vigo
Publication: SIAM Review
Problem: Efficiently managing multiple delivery routes is a complex challenge due to constraints like vehicle capacity, time windows, and delivery locations.
Methodology: The study reviews various optimization techniques including heuristics, metaheuristics, and exact algorithms to solve the Vehicle Routing Problem.
Result: Advanced optimization techniques significantly improved route efficiency and reduced operational costs in logistics systems.
Inference: This research supports GrabGo's delivery optimization module, enabling efficient route planning and improved delivery performance.

2.11 Shortest Path Optimization

Title: Implementation of Dijkstra Algorithm in Determining the Fastest Route for Goods Delivery
Author: Implementation of Dijkstra Algorithm in Determining the Fastest Route for Goods Delivery
Publication: International Journal of Advanced Computer Science (2025)
Problem: Manual route selection by delivery personnel often leads to longer travel distances and delays in delivery.
Methodology: The study applied Dijkstra's algorithm to calculate the shortest path in a weighted graph representing delivery routes.
Result: The system achieved a 90% success rate in selecting optimal routes and reduced delivery distance by 8.55**Inference:**GrabGo uses Dijkstra-based route optimization to ensure faster deliveries and reduced transportation costs.

2.12 Sentiment Analysis for Reviews

Title: Sentiment Analysis and Opinion Mining
Author: Bing Liu
Publication: Morgan Claypool Publishers
Problem: Understanding customer opinions from textual reviews is challenging due to unstructured data and varying expressions.
Methodology: The research introduces NLP-based sentiment analysis techniques to classify reviews as positive, negative, or neutral.
Result: Sentiment analysis improved the understanding of customer feedback and enhanced decision-making processes.

Inference: GrabGo integrates sentiment analysis to evaluate user feedback and improve recommendation and review filtering systems.

2.13 A Survey of Recommender Systems

Title:A Survey of Recommender Systems
Author: Francesco Ricci, Lior Rokach, Bracha Shapira
Publication: Springer
Problem: Users face difficulty in finding relevant products due to the large volume of available options in online platforms.
Methodology: The study reviews collaborative filtering, content-based filtering, and hybrid recommendation approaches.
Result: Hybrid recommendation systems significantly improved recommendation accuracy and user satisfaction.
Inference: GrabGo uses hybrid recommendation techniques to provide personalized product suggestions and enhance user experience.

2.14 NLP-Based Review Spam Detection

Title: Detecting Opinion Spam Using Natural Language Processing
Author: Jindal and Liu
Publication: Proceedings of WWW Conference
Problem: Fake reviews mislead customers and reduce trust in online platforms.
Methodology: The study used NLP techniques and supervised learning models to detect spam reviews based on linguistic patterns and behavioral features.
Result: The model successfully identified spam reviews with high accuracy using textual and behavioral analysis.
Inference: GrabGo applies NLP-based spam detection techniques to automatically identify and filter fake reviews, ensuring platform credibility.

Table 2.1: Comparison of Existing Systems and GrabGo Solution

Title of the Research Paper	**Gaps / Limitations Identified**	**How GrabGo Will Fill the Gap**
A Predictive–Reactive Optimization Framework for Food Delivery	Focuses mainly on delivery scheduling and routing efficiency but does not address integration with small-scale vendor platforms or user-friendly mobile applications. Limited emphasis on real-time customer interaction features.	GrabGo integrates predictive delivery optimization with a complete multi-user platform including Customer, Vendor, Delivery, and Admin applications. It also provides real-time order tracking, notifications, and a simple mobile interface for better usability.
Reinforcement Learning-Based AI-Powered Route Optimization	The system focuses on route optimization but lacks integration with order management, vendor coordination, and customer communication systems. It mainly addresses logistics efficiency.	GrabGo combines route optimization with full delivery management features such as order processing, delivery tracking, vendor coordination, and real-time updates to improve overall system performance.
Smart Chatbot System for E-Commerce Assistance based on AIML	The chatbot system handles customer queries but lacks advanced personalization, integration with delivery tracking, and multi-role system support. It is limited to basic customer service functions.	GrabGo plans to implement an intelligent chatbot integrated with the delivery system to handle order queries, delivery status updates, product recommendations, and automated customer support.
Towards Understanding and Detecting Fake Reviews in App Stores	The system detects fake reviews but does not integrate the detection mechanism directly into real-time e-commerce or delivery platforms. It focuses only on review analysis.	GrabGo introduces a Trust Filter system that monitors review patterns, detects suspicious activity, and ensures reliable vendor ratings to maintain platform transparency and customer trust.
E-commerce Personalized Recommendations Based on Deep Neural Collaborative Filtering	The recommendation model focuses on user preferences but does not consider real-time delivery factors such as availability, distance, or delivery time.It mainly targets product recommendation accuracy	GrabGo enhances recommendation systems by combining user preferences with real-time factors like product availability, location, and delivery time to provide smarter and more relevant suggestions.
Ensemble Learning-Based Prediction Model for Enhanced Marketing and Recommendation	The model improves prediction accuracy but lacks implementation in small-scale delivery platforms and does not address demand forecasting for local vendors.	GrabGo uses ensemble learning techniques to predict product demand and customer behavior, helping vendors manage inventory efficiently and improve service quality
Trust and Online Purchase Intention: A Systematic Literature Review through Meta-Analysis	The research identifies the importance of trust but does not provide a direct technical implementation method for building trust in delivery platforms	GrabGo implements secure payment systems, verified vendor profiles, transparent review mechanisms, and clear policies to increase customer trust and encourage repeat purchases.
Cloud-Based Food Ordering Platform	Limited scalability and personalization features	GrabGo uses scalable cloud infrastructure with personalized suggestions

Title of the Research Paper	**Gaps / Limitations Identified**	**How GrabGo Will Fill the Gap**
Sales Demand Forecasting Using Long Short-Term Memory (LSTM) Neural Networks	The study focuses only on time-series demand prediction and does not integrate demand forecasting with delivery resource allocation or vendor inventory management. It also requires large datasets and high computational resources.	GrabGo will use lightweight LSTM-based models integrated with vendor inventory and delivery scheduling systems. This ensures accurate demand prediction while optimizing delivery staff allocation and reducing operational costs.
The Vehicle Routing Problem: A Survey	The research provides theoretical optimization techniques but does not address real-time traffic updates, sudden delivery changes, or dynamic route re-planning during delivery operations.	GrabGo will implement dynamic route optimization algorithms that continuously update delivery routes using real-time traffic data, GPS tracking, and order priority levels. This ensures faster delivery and efficient vehicle utilization.
Implementation of Dijkstra Algorithm in Determining the Fastest Route for Goods Delivery	The Dijkstra algorithm identifies the shortest path but does not consider traffic congestion, road closures, weather conditions, or multiple delivery destinations simultaneously.	GrabGo will combine Dijkstra's algorithm with real-time traffic and multi-stop route optimization, enabling the system to calculate the fastest and most efficient delivery path dynamically.
Sentiment Analysis and Opinion Mining	The study focuses on classifying customer reviews but does not integrate sentiment results into recommendation systems or vendor performance evaluation.	GrabGo will integrate sentiment analysis with recommendation and vendor rating systems, allowing the platform to automatically identify customer satisfaction levels and improve product recommendations.
A Survey of Recommender Systems	The research reviews recommendation techniques but does not address real-time personalization based on user behavior, location, or purchase history. It also lacks integration with delivery systems.	GrabGo will implement AI-driven personalized recommendation systems that analyze user preferences, order history, location, and browsing behavior to provide accurate product suggestions in real time
Detecting Opinion Spam Using Natural Language Processing	The system focuses primarily on text-based spam detection and does not consider behavioral patterns such as unusual review frequency or suspicious account activity.	GrabGo will implement a hybrid fake review detection system that combines NLP-based text analysis with behavioral monitoring to identify fraudulent reviews and maintain platform trust.

Chapter 3

Software Requirements Specification

3.1 Introduction

The Software Requirements Specification (SRS) outlines the functional and non-functional requirements for the proposed GrabGo system. GrabGo is an AI-driven, all-in-one delivery intelligence platform that includes three connected applications: a Customer Application, a Vendor Application, and an Admin Application. These applications work together through a centralized backend to provide personalized recommendations, review analysis, delivery optimization, real-time customer support, and overall system management. The SRS serves as a guide for system design, development, testing, and deployment.

3.2 Functional requirements

The functional requirements of the GrabGo system are divided into three applications.

3.2.1 Customer Application

The system will allow customers to register and log in securely. It will enable customers to browse products in various categories. The system will provide personalized product and seller recommendations while showing real reviews and filtering out fake or biased ones. Customers will be able to place orders and track deliveries in real time. The system will predict and display accurate delivery times. It will include an AI-driven chatbot to provide immediate responses to customer questions about products, orders, and delivery status.

3.2.2 Vendor Application

The system will allow vendors to register and log in securely. It will enable vendors to add, update, and manage their product listings. Vendors will be able to set prices and manage stock availability. The system will provide vendors with customer feedback and insights on their performance. It will also allow vendors to process and manage incoming orders.

3.2.3 Admin Application

The system will enable administrators to manage users, vendors, and product categories. It will monitor performance and activity on the platform. The system will oversee review moderation and model performance. It will also control platform policies, analytics, and overall configuration.

3.3 Non-Functional requirements

Non-functional requirements describe the quality attributes, constraints, and performance expectations of the GrabGo system. These requirements ensure the system operates efficiently, securely, and reliably. They also help provide a smooth user experience across the Customer, Vendor, and Admin applications.

3.3.1 Performance Requirements

The system will offer quick and responsive performance to ensure users have a smooth experience. All user actions, like browsing products, submitting reviews, placing orders, tracking deliveries, and interacting with the chatbot, should be processed promptly. The system will efficiently manage many users simultaneously without delays. Delivery route optimization and recommendation calculations will be handled with minimal delays to support real-time decision making.

3.3.2 Security and Reliability Requirements

The system will ensure secure handling of user and vendor data through proper authentication, authorization, and data validation methods. Sensitive information, like user credentials and transaction details, will be protected from unauthorized access. The system must be reliable and maintain steady availability so services remain accessible even during peak usage. The fake review detection mechanism will operate accurately and consistently. Administrators will monitor system outputs to maintain fairness and transparency.

3.4 User Interface requirements

The user interface requirements outline how users interact with the GrabGo system. The system consists of three applications: Customer, Vendor, and Admin. The interfaces must be easy to use, consistent, and responsive to support efficient interaction. The goal of the user interface design is to simplify navigation, reduce user effort, and improve overall usability. Clear visual elements, logical workflows, and real-time feedback are essential to provide a smooth and engaging user experience across all applications.

3.4.1 User Interface Design Requirements

The user interface will feature a clean, consistent, and organized design across all three applications. The Customer application will provide easy access to product categories, personalized recommendations, reviews, order placement, delivery tracking, and the AI-driven chatbot. The Vendor application will feature dashboards for managing product listings, prices, orders, and performance feedback. The Admin application will display monitoring interfaces for managing users, vendors, reviews, and system analytics. All interfaces will be responsive and adjustable to different screen sizes, supporting both web and mobile platforms.

3.4.2 Usability and Accessibility Requirements

The user interface will be designed to be easy to use and accessible to users with varying levels of technical knowledge. Navigation will be straightforward, with minimal steps needed to complete common tasks like browsing products, placing orders, or managing listings. We will use readable fonts, good color contrast, and clear icons to enhance visibility and user comfort. The system will provide helpful notifications, error messages, and confirmations to effectively guide users. Additionally, the AI-driven chatbot will assist users by offering real-time support and answering questions, further enhancing usability.

3.5 Software Requirements

The software requirements outline the programming languages, frameworks, tools, and platforms needed for developing and deploying the GrabGo system. The system consists of three connected applications: Customer, Vendor, and Admin, all supported by a centralized backend and database. The chosen software technologies ensure compatibility across platforms, scalability, security, and the effective integration of machine learning, natural language processing, and real-time services.

3.5.1 Application Development Software

Flutter will be used to build the front end of the GrabGo system. Flutter allows you to create apps for both web and mobile using the same codebase. It makes it easy to develop a responsive and high-performance user interface for the Customer, Vendor, and Admin apps. The Dart language will be used for the front end. Python will power the backend of the system. Python supports rapid development and works well with machine learning and NLP libraries. A backend framework like FastAPI will handle user requests, authentication, product management, order processing, and more by creating RESTful API interactions with chatbots. FastAPI simplifies the documentation of APIs, speeds up request handling, and manages requests asynchronously.

3.5.2 Database, Machine Learning, and Supporting Tools

The system will use SQLite as the database to store user profiles, vendor information, product details, reviews, orders, and system logs. SQLite is a lightweight database solution suitable for prototype and academic project development. The database design will be scalable to accommodate other database systems if needed. Machine learning and natural language processing components such as recommendation systems, counterfeit review detection, delivery time prediction, and the AI-powered chatbot will be developed using Python libraries. Libraries like NumPy, Pandas, and Scikit-learn will help with data processing and modeling, while NLP libraries can be used for text analysis and chatbot response generation. Development and testing will utilize tools like Visual Studio Code as the Integrated Development Environment (IDE). API testing tools will be used for testing backend services, while web browsers or actual mobile devices will serve for application testing. The system will be compatible with standard operating systems like Windows or macOS.

3.6 Hardware Requirements

The hardware requirements include the minimum system configuration needed to develop, test, and run the GrabGo platform. The GrabGo platform requires a system that can support application development, backend execution, and machine learning. Client-side users need standard devices such as smartphones and computers with internet access. The hardware configuration should run all three applications without any degradation.

Chapter 4

System Design

4.1 Architecture Design

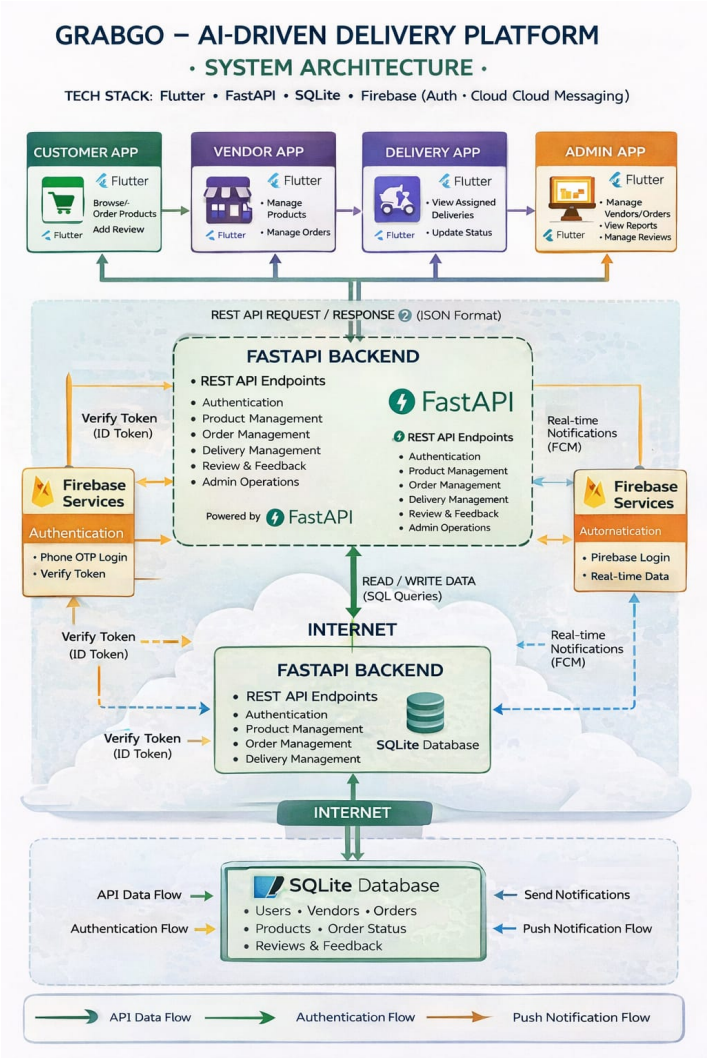

Figure 4.1: System Architecture Diagram

This Figure 5.1,the illustration represents the system architecture of the GrabGo AI-Driven Delivery Platform, which consists of four main applications: Customer, Vendor,

Delivery, and Admin. All applications are developed using Flutter and communicate with a centralized backend built using FastAPI through REST API requests in JSON format. The system uses Firebase for secure user authentication using OTP and for sending real-time notifications to users and delivery agents.

The backend processes requests such as product management, order handling, and delivery tracking, and stores all system data in a structured database using SQLite. This architecture ensures secure communication, efficient data management, and real-time updates across all applications, making the platform reliable and scalable for managing delivery operations.

4.2 Decomposition Description

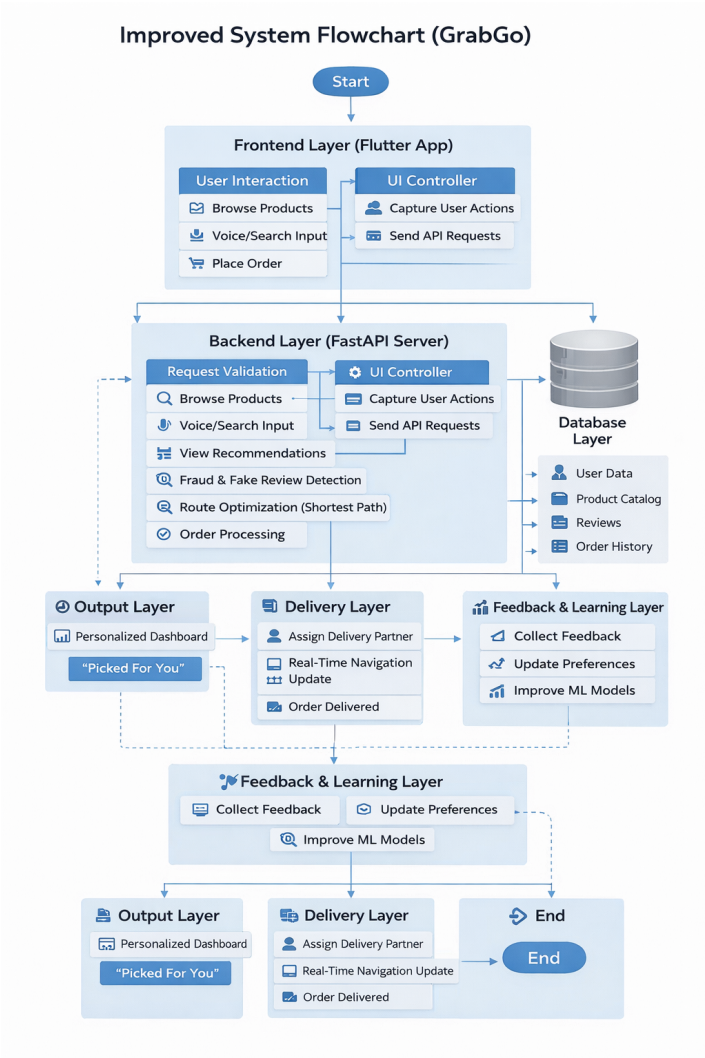

Figure 4.2: Flow chart

Figure 5.2 illustrates that GrabGo system is decomposed into four main application modules: Customer App, Vendor App, Delivery App, and Admin App, each responsible for specific user operations such as ordering products, managing inventory, handling deliveries, and monitoring system activities. All modules are developed using Flutter and communicate with a centralized backend through secure API requests.

The backend module, implemented using FastAPI, manages core functionalities including authentication, order processing, and delivery management, while data is stored in a database using SQLite. User authentication and real-time notifications are handled by Firebase, ensuring secure access and timely communication across the entire system.

4.3 Data Flow Design

Data Flow Design The data flow in the GrabGo system begins when users interact with the Customer, Vendor, Delivery, or Admin applications developed using Flutter. User actions such as placing orders, updating delivery status, or managing products are sent as API requests to the backend server built using FastAPI. The backend processes these requests, performs the required operations, and communicates with the database to store or retrieve information.

All system data, including users, products, orders, and delivery status, is stored and managed using SQLite. Secure login and user verification are handled through Firebase authentication, while real-time notifications are sent to users and delivery agents to update them about order status and system events. This structured data flow ensures reliable communication, data consistency, and efficient system performance.

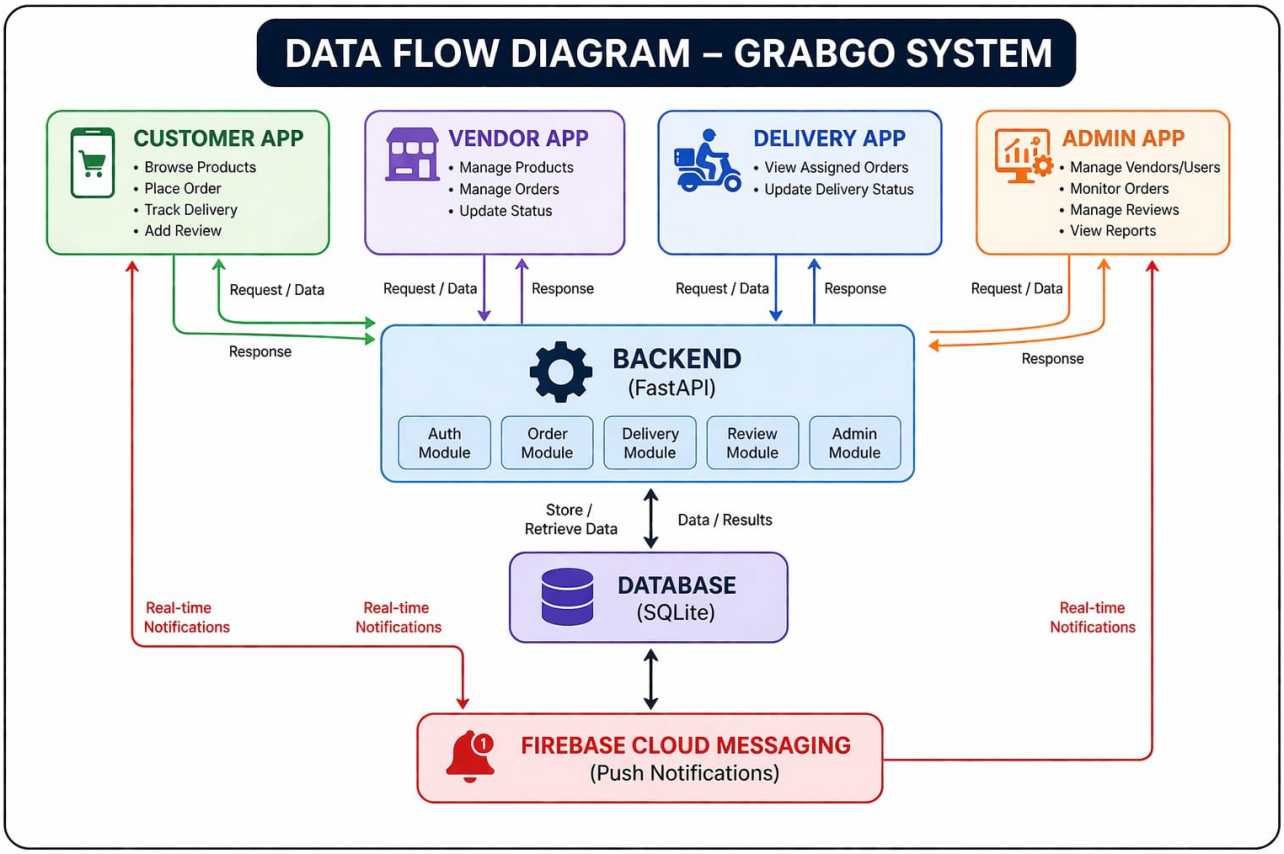

Figure 4.3: Dataflow design

4.4 Use case Diagram

The Use Case Diagram for the GrabGo system represents the interaction between different users (actors) and the system functionalities. It shows how each user communicates with the system to perform specific operations related to ordering, selling, delivering, and managing products. The main actors in the system are Customer, Vendor, Delivery Agent, and Admin. Each actor has defined responsibilities that ensure the smooth functioning of the delivery platform.

The system is implemented using modern technologies such as Flutter for developing mobile applications, FastAPI for handling backend services and APIs, SQLite for storing system data, and Firebase for authentication and notifications. The Use Case Diagram helps developers and stakeholders understand system requirements, user roles, and system functionality clearly.

Actors in the Use Case Diagram:

Customer: The Customer is the primary user who interacts with the system to purchase products or services.

Vendor: The Vendor is responsible for managing products and responding to customer orders.

Vendor Use Cases

. Register vendor account
. Login to system
. Add new products
. Update product details
. Delete products
. Manage inventory
. View customer orders
. Accept or reject orders
. Update product availability
. View sales reports
. Receive notifications

The vendor ensures that products are available and orders are processed efficiently.

Delivery Agent: The Delivery Agent is responsible for delivering orders to customers.

Delivery Agent Use Cases
. Register delivery account
. Login to system
. View assigned deliveries
. Accept delivery request
. Navigate to delivery location
. Update delivery status
. Confirm order delivery
. Contact customer
. Receive delivery notifications

The delivery agent ensures timely and accurate delivery of products.

Extract Features: This use case involves extracting relevant features from the integrated audio-visual representation. It may include computing spectrograms, MFCCs, facial landmarks, or other visual and audio features.

Perform Speech Separation: This use case focuses on the actual speech separation process. It utilizes the extracted audio and visual features to separate the individual speech signals from the mixture, using techniques such as blind source separation or deep learning-based models.

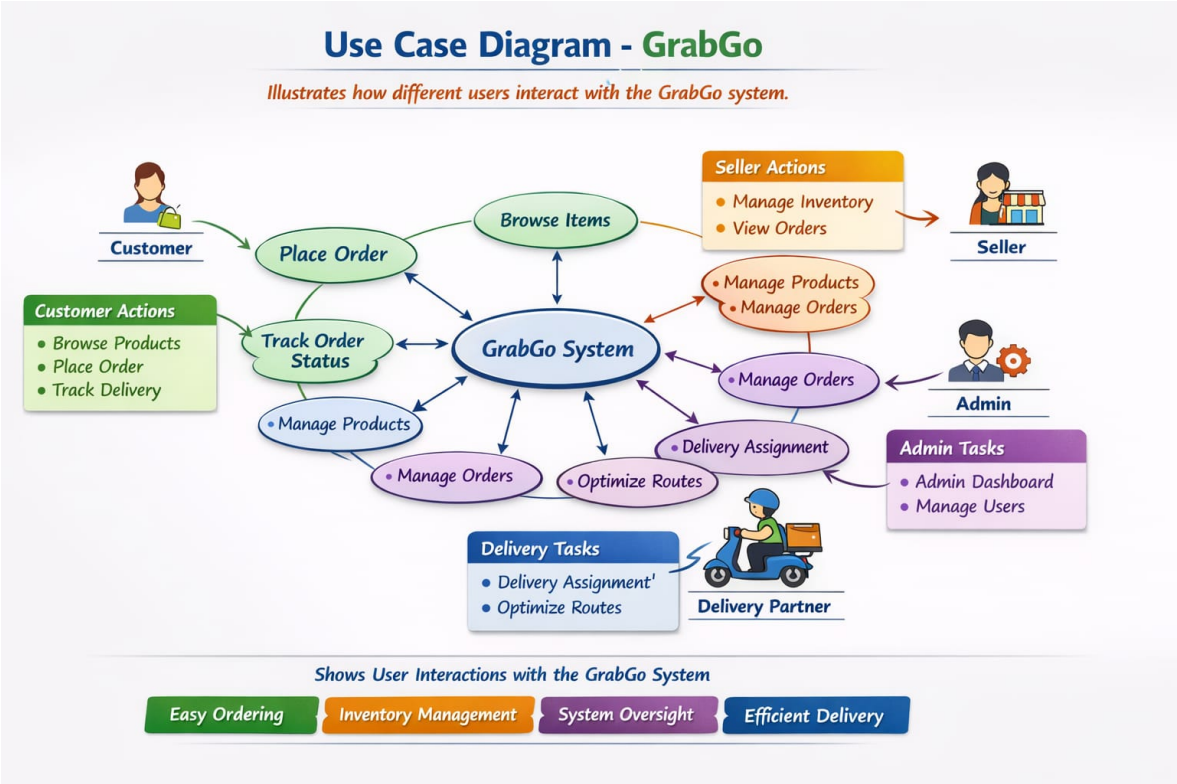

Figure 4.4: Use case diagram for Grabgo system

Chapter 5

Implementation

5.1 Audio Extraction

Audio extraction is the process of isolating and extracting the audio content from a multimedia source, such as a video file. It involves separating the audio track from the accompanying video or other elements to obtain a standalone audio file representing the sound present in the source material.

Figure 5.1: code snippet for audio extraction

5.2 Speech Separation

SpeechBrain is an open-source framework

5.2.1 Sepformer

SepFormer is an algorithm for speech separation that utilizes self-attention mechanisms. It employs a transformer-based architecture to capture long-range dependencies and

model the relationships between time-frequency points in the audio mixture, enabling the separation of multiple speech sources from the mixture.

Figure 5.2: code snippet for speech separation

5.3 Speech Enhancement

5.3.1 Lite Audio Visual Speech Enhancement

Lite AVSE algorithm is used for the separation and enhancement of the speech. The system includes two visual data compression techniques and removes the visual feature extraction network from the training model, yielding better online computation efficiency. As for the audio features, short-time Fourier transform (STFT) is calculated of 3-second audio segments. Each time-frequency (TF) bin contains the real and imaginary parts of a complex number, both of which used as input. Power-law compression used to prevent loud audio from overwhelming soft audio. The same processing is applied to both the noisy signal and the clean reference signal.

5.3.2 Spectral Subtraction

Spectral subtraction is a technique used in audio signal processing to reduce background noise from an audio signal. It involves estimating the noise spectrum from a noisy signal and subtracting it from the noisy spectrum to enhance the desired signal. The resulting spectrum is then transformed back into the time domain to obtain a cleaner audio signal.

Figure 5.3: code snippet for speech enhancement using LAVSE

Figure 5.4: code snippet for speech enhancement using spectral subtraction

5.4 Speaker Detection

The cv2 functions provide methods to load the pre-trained models, apply them to images or video frames, and draw bounding boxes around the detected faces. By leveraging cv2's face detection capabilities, you can automate tasks such as facial recognition, emotion analysis, or face tracking in various applications like surveillance, biometrics, or augmented reality.

Figure 5.5: code snippet for speaker detection

Chapter 6

System Testing

Testing is a procedure of executing the program with unequivocal intension of [1] discovering mistakes, assuming any, which makes the program, fall flat. This stage is an essential piece of improvement.

It plays out an exceptionally basic part for quality affirmation and for guaranteeing unwavering quality of programming. It is the way toward finding the mistakes and missing operation and furthermore an entire confirmation to decide if the targets are met the client prerequisites are fulfilled.

The objective of testing is to reveal prerequisites, outline or coding blunders in the projects. Therefore, unique levels of testing are utilized in programming frameworks. The testing results are utilized amid upkeep.

This area manages the points of interest in the various classes of the test which should be directed to approve capacities, imperatives and execution. This can be accomplished fundamentally by using the methods for testing, which assumes a crucial part in the improvement of a product.

The structure of the program is not being considered in useful testing. Test cases are exclusively chosen on the premise of the prerequisites or particulars of a program or module of program but the internals of the module or the program are not considered for determination of experiments.

The program to be tried is executed with an arrangement of experiments and the yield of the program for the experiments is assessed to decide whether the program is executing not surprisingly. The accomplishment of testing in uncovering mistakes in projects depends basically on the experiments. There are two fundamental ways to deal with testing Black Box or functional Testing and White Box or structural testing.

Table 6.1: Work Flow

Sl No	Work	Duration(in Weeks)
1	Audio Extraction	1
2	Audio Enhancement using LAVSE	4
3	Audio Separation using Speechbrain	3
4	Noice Reduction using spectral subtraction	2
5	Image segmentation	3
6	Speaker Identification	5

Figure 6.1: testcases

Chapter 7

Results and Discussion

7.1 Face detection

Figure 7.1: Face detection

Above figure 8.1 shows initial face detection process using opencv and dlib. It convert the image to grayscale, apply the model using cv2.detectMultiScale(), and draw bounding boxes around the detected faces using cv2.rectangle(). Display or save the result using cv2.imshow() or cv2.imwrite().

7.2 Speaker recognition

Figure 7.2: Speaker recognition 1,person 1

Figure 7.3: Speaker recognition 1,person 2

Figure 7.4: Speaker recognition 2,person 1

Figure 7.5: Speaker recognition 2,person 2

Figure 7.6: Speaker recognition 3,person 1

Figure 7.7: Speaker recognition 3,person 2

Figures from 8.2 to 8.7 shows speaker recognition process using opencv and dlib.Speaker detection using cv2 and dlib involves utilizing dlib's pre-trained models along with cv2 functions to detect and locate human faces. By combining face detection with additional techniques such as audio analysis or lip movement tracking, speaker detection can be achieved in various applications like video conferencing or surveillance. [2] [7]

Chapter 8

Conclusion and Future work

The Project will help in narrowing the imprecise communication problem in real-time data using speech separation and speaker identification technique by Deep Learning and Image Processing algorithms. This will impact the communication and security sectors in a greater extent. Overall, this project aims to develop an application or method that can help to separate the audio-visual speech and enhance it based on speaker identification.
This project can be further developed as:

- By incorporating more real-world testing and gathering feedback from individual units.
- The system can be connected with communication devices or services to enable the users to communicate with others with ease.

This project has a great potential to make a positive impact on communication and security situations. Its continuous improvement will be important to make this impact even greater [9]

Example for citing and bibtex for journal paper [3], conference paper [8], citing website [5], citing book [4]

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Appendix A

Screenshots

Figure A.1: Call this figure A.1.

Figure A.2: Call this figure A.2.

Appendix B

User Manual

Add the contents here